NOETHERIAN APPROXIMATION PRESERVING GEOMETRIC IRREDUCIBILITY OF THE FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 9879-9886 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 9888-9912, the complete author proof (23 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Noetherian approximation of finitely presented morphisms of affine schemes and geometric irreducibility of fibres; the descent is retained.

Source reference. Original lemma lemma-Noetherian-approximation: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9739–9753. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-base-change-fibres-geometrically-irreducible: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7432–7440. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-nr-geom-irreducible-components-constructible: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7794–7804. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05F4: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9912–9912. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a morphism of affine schemes, which is of finite presentation with geometrically irreducible fibres. Then there exists a diagram as in Lemma lemma-Noetherian-approximation such that in addition f_0 has geometrically irreducible fibres.*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Apply Lemma lemma-Noetherian-approximation to get a cartesian diagram

$$\begin{array}{ccc} X_0 & \xleftarrow{g} & X \\ {\scriptstyle f_0}\downarrow & & \downarrow{\scriptstyle f} \\ S_0 & \xleftarrow{h} & S \end{array}$$

of affine schemes with $X_0 \to S_0$ a finite type morphism of schemes of finite type over $\mathbf{Z}$. By Lemma lemma-nr-geom-irreducible-components-constructible the set $E \subset S_0$ of points where the fibre of f_0 is geometrically irreducible is a constructible subset. By Lemma lemma-base-change-fibres-geometrically-irreducible we have $h(S) \subset E$. Write $S_0 = \mathrm{Spec}(A_0)$ and $S = \mathrm{Spec}(A)$. Write $A = \mathrm{colim}_i A_i$ as a direct colimit of finite type A_0-algebras. By Limits, Lemma 05F4 we see that $\mathrm{Spec}(A_i) \to S_0$ has image contained in E for some i. After replacing S_0 by $\mathrm{Spec}(A_i)$ and X_0 by $X_0 \times_{S_0} \mathrm{Spec}(A_i)$ we see that all fibres of f_0 are geometrically irreducible. $\square$